Introduction to Artificial Intelligence and Machine Learning (CSN11601)

Assignment-5

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Q1. Create a NumPy array and apply methods such as append(), insert(), pop(), remove(), index(), sort(), and reverse().

A1. CODE:

```
import numpy as np
import numpy as np
arr = np.array([5, 2, 8, 1])
print("Original:", arr)
arr = np.append(arr, 9)
print("Append:", arr)
arr = np.insert(arr, 2, 7)
print("Insert:", arr)
arr = np.delete(arr, 1)
print("Pop:", arr)
arr = np.delete(arr, np.where(arr == 8))
print("Remove 8:", arr)
print("Index of 7:", np.where(arr == 7)[0][0])
arr.sort()
print("Sorted:", arr)
arr = arr[::-1]
print("Reversed:", arr)
```

OUTPUT:

```
p(base) user@wipro-desktop:~/Desktop/Aayansh_20253004/Assignment-5$ python p1.py
Original: [5 2 8 1]
Append: [5 2 8 1 9]
Insert: [5 2 7 8 1 9]
Pop: [5 7 8 1 9]
Remove 8: [5 7 1 9]
Index of 7: 1
Sorted: [1 5 7 9]
Reversed: [9 7 5 1]
```

Q2. Write a program to copy array with assignment operator, shallow copy method and deep copy method.

A2. CODE:

```
import numpy as np

a = np.array([1, 2, 3])
b = a
c = a.view()
d = a.copy()
a[0] = 99
print("a:", a)
print("b (assign):", b)
print("c (shallow):", c)
print("d (deep):", d)
```

OUTPUT:

```
(base) user@wipro-desktop:~/Desktop/Aayansh_20253004/Assignment-5$ python p2.py
a: [99  2  3]
b (assign): [99  2  3]
c (shallow): [99  2  3]
d (deep): [1 2 3]
```

Q3. Write a program for math and cmath module functionality in your program. E.g., math.acos(), math.ceil(), math.exp(), math.gcd(), cmath.tan(x), cmath.log10(x), cmath.isclose().

A3. CODE:

```
import math, cmath
print(math.acos(1))
print(math.ceil(4.3))
print(math.exp(2))
print(math.gcd(12, 8))
x = 3 + 4j
print(cmath.tan(x))
print(cmath.log10(x))
print(cmath.isclose(1.0001, 1.0002))
```

OUTPUT:

```
(base) user@wipro-desktop:~/Desktop/Aayansh_20253004/Assignment-5$ python p3.py
0.0
5
7.38905609893065
4
(-0.00018734620462947 84+0.999355987381473j)
(0.6989700043360187+0.4027191962733731j)
False
```

Q4. Write a program to create a calculator using function.
A4. CODE:

```
def add(a,b): return a+b
def sub(a,b): return a-b
def mul(a,b): return a*b
def div(a,b): return a/b
a = float(input("Enter a: "))
b = float(input("Enter b: "))
op = input("Op (+ - * /): ")
if op == '+': print(add(a,b))
elif op == '-': print(sub(a,b))
elif op == '*': print(mul(a,b))
elif op == '/': print(div(a,b))
else: print("Invalid")
```

OUTPUT:

```
(base) user@wipro-desktop:~/Desktop/Aayansh_20253004/Assignment-5$ python p4.py
Enter a: 123
Enter b: 321
Op (+ - * /): +
444.0
(base) user@wipro-desktop:~/Desktop/Aayansh_20253004/Assignment-5$ python p4.py
Enter a: 342
Enter b: 211
Op (+ - * /): -
131.0
(base) user@wipro-desktop:~/Desktop/Aayansh_20253004/Assignment-5$ python p4.py
Enter a: 12
Enter b: 4
Op (+ - * /): *
48.0
(base) user@wipro-desktop:~/Desktop/Aayansh_20253004/Assignment-5$ python p4.py
Enter a: 12
Enter b: 4
Op (+ - * /): /
3.0
```